

Intrinsic interval consequences of Gern's type D thesis

A research note using combinatorial invariance

Research draft prepared with an Astra research team

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Abstract

We extract consequences of Gern's thesis that apply to arbitrary equal-parameter Coxeter systems by using Theorem 1.1 of the September 24, 2026 combinatorial invariance preprint. A type D interval with fully commutative bottom becomes a certificate for a 0–1 leading coefficient in every realization of its abstract interval poset. More specifically, Gern's bad-element intervals give an infinite family of degree bounds and exact leading coefficients, including vanishing of increasingly many top coefficients. All these statements are conditional on the cited invariance theorem; we do not independently certify its full proof. We distinguish these formal consequences from previously known type B bounds, parabolic-coset transfers, and subsystem pattern embeddings. No new whole-type theorem or publication-level novelty claim is established here.

1 Inputs and the precise scope of transfer

Throughout, $P_{u,v}(q)$ is the ordinary equal-parameter Kazhdan–Lusztig polynomial of a Coxeter system (W, S) . We use

$$P_{u,u} = 1, \quad P_{u,v} = 0 \ (u \not\leq v), \quad \deg P_{u,v} \leq \left\lfloor \frac{\ell(v) - \ell(u) - 1}{2} \right\rfloor \quad (u < v).$$

For $u < v$ put $d = \ell(v) - \ell(u)$ and define

$$\mu(u, v) = \begin{cases} [q^{(d-1)/2}]P_{u,v}(q), & d \text{ odd}, \\ 0, & d \text{ even}. \end{cases}$$

For odd d , μ is the coefficient at the largest *allowed* degree; it need not be the leading nonzero coefficient. For even d , μ is zero by convention.

The two external inputs are:

1. Gern [1, Theorem 4.5.11]: in finite type D_n , if x is fully commutative and $x < w$, then $\mu(x, w) \in \{0, 1\}$.
2. OpenAI [2, Theorem 1.1]: for arbitrary Coxeter systems, an order isomorphism $[u, v] \cong [x, w]$ preserves the ordinary equal-parameter polynomial: $P_{u,v} = P_{x,w}$.

The second statement includes infinite and noncrystallographic systems. It requires an isomorphism of the *whole closed interval*. An embedding, equal rank numbers, an undirected graph match, or a reflection-subgroup inclusion is not sufficient. The consequences proved below invoke this input, rather than silently treating the preprint as an independently verified theorem.

Definition 1.1. Let $\mathcal{C}_D$ be the class of finite posets isomorphic to an interval $[x, w]$ in some D_n , $n \geq 4$, with x fully commutative. A witness consists of n, x, w and an explicit order isomorphism.

Theorem 1.2 (Type D witness certificate). *Assume the invariance input. In any Coxeter system, if $[u, v] \in \mathcal{C}_D$ and $u < v$, then $\mu(u, v) \in \{0, 1\}$. The element u is not required to be fully commutative.*

Proof. Choose a witness $\iota : [u, v] \rightarrow [x, w]$. Bruhat intervals are finite graded posets, and an order isomorphism preserves their rank. Hence $\ell(v) - \ell(u) = \ell(w) - \ell(x)$. The invariance input gives $P_{u,v} = P_{x,w}$. For odd rank the coefficients defining μ are consequently equal; for even rank both are zero. Gern's theorem applies to the witness bottom x . $\square$

Corollary 1.3 (Obstruction). *Assume the invariance input. If $\mu(u, v) \geq 2$, then $[u, v]$ has no fully commutative-bottom type D witness.*

Proof. This is the contrapositive of Theorem 1.2. $\square$

This certificate theorem is an immediate consequence of the two inputs, not a classification of $\mathcal{C}_D$ or an independent new mechanism. In particular, it does not prove the fully commutative-bottom conjecture throughout an exceptional or affine type: that would require witnesses for *all* the relevant intervals or a separate argument.

Proposition 1.4 (Fully commutative status is not encoded across types). *The abstract principal lower interval does not determine whether its top is fully commutative across arbitrary Coxeter systems.*

Proof. Take s, t with $m(s, t) = 3$, and separately s', t' with $m(s', t') = 4$. The intervals $[e, sts]$ and $[e, s't's']$ each have levels

$$\{e\}, \quad \{s, t\}, \quad \{st, ts\}, \quad \{sts\},$$

with both rank-one elements below both rank-two elements (and the corresponding primed levels in the second interval). This follows directly from the subword criterion. They are therefore isomorphic six-element posets. In $I_2(3)$ the braid $sts = tst$ makes sts non-fully-commutative. In $I_2(4)$ a reduced word of length three cannot contain the noncommuting braid of length four; $s't's'$ is fully commutative. See also Stembridge's definition and braid characterization [5]. $\square$

2 An infinite family carrying the thesis's exact calculations

We use Gern's numbering: s_1 sends the first two signed entries (a, b) to $(-b, -a)$, s_2 interchanges the first two entries, and s_i interchanges positions $i - 1, i$ for $i \geq 3$. Right multiplication has these effects on signed one-line notation. For even $n \geq 4$ define

$$x_n = s_1 s_2 s_4 s_6 \cdots s_n, \quad w_n(i) = \begin{cases} (-1)^{n/2}, & i = 1, \\ i, & i > 1 \text{ odd}, \\ -(n + 2 - i), & i \text{ even}. \end{cases}$$

Set $I_n = [x_n, w_n]$. These are Gern's Definition 2.3.8 and Corollary 2.2.19. The factors of x_n commute, so x_n is fully commutative. The formulas also give an even number of negative entries in w_n , as required for D_n . For odd $n \geq 5$, the corresponding thesis elements are the even $(n - 1)$ elements with an additional fixed last entry; their intervals are identical after standard parabolic inclusion.

The following exact inputs are from the thesis:

$$L_n := \ell(w_n) = \frac{3n^2}{8} + \frac{n}{4}, \quad \ell(x_n) = \frac{n}{2} + 1, \quad (1)$$

$$d_n := \text{rank } I_n = \frac{3n^2}{8} - \frac{n}{4} - 1, \quad (2)$$

$$a_n := a(w_n) = \begin{cases} 3n/4, & n \equiv 0 \pmod{4}, \\ (3n+2)/4, & n \equiv 2 \pmod{4}, \end{cases} \quad (3)$$

$$P_{x_n, w_n} = P_{e, w_n}, \quad (4)$$

$$\mu(x_n, w_n) = \begin{cases} 1, & n = 6, \\ 0, & n \neq 6. \end{cases} \quad (5)$$

The locators are Lemma 2.2.20, Definition 2.3.8, Proposition 4.5.1, Lemma 2.3.9, and Lemmas 4.5.2–4.5.7, respectively. Equation (5) uses the explicit computer calculations at $n = 6$ and $n = 8$ recorded in Lemma 4.5.5; it is not entirely a symbolic argument in the original thesis. The thesis's Proposition 1.3.5, citing Lusztig [9, 1.3(a)], gives

$$\deg P_{e, w_n} \leq \frac{L_n - a_n}{2}.$$

Theorem 2.1 (Bad-interval degree and coefficient transfer). *Assume the invariance input. Let $[u, v]$ be an interval in any Coxeter system, and suppose $[u, v] \cong I_n$ for even $n \geq 4$. Put*

$$M_n = \left\lfloor \frac{d_n - 1}{2} \right\rfloor, \quad A_n = \left\lfloor \frac{L_n - a_n}{2} \right\rfloor, \quad B_n = \min\{M_n, A_n\}.$$

Then

$$\mu(u, v) = \begin{cases} 1, & n = 6, \\ 0, & n \neq 6, \end{cases} \quad \deg P_{u, v} \leq B_n.$$

For $n = 8$ the bound improves to $\deg P_{u, v} \leq 9$.

Proof. Rank preservation identifies the rank with d_n , and invariance gives $P_{u, v} = P_{x_n, w_n} = P_{e, w_n}$. The exact μ assertion follows from (5). The ordinary degree bound gives M_n , and Lusztig's bound with (3) gives A_n , taking floors because polynomial degrees are integral. When $n = 8$, $d_8 = 21$ and $M_8 = A_8 = 10$. Equation (5) says the coefficient of q^{10} vanishes. Thus the degree is at most nine. $\square$

Corollary 2.2 (Increasing gaps at the allowed top degree). *Assume the invariance input and $[u, v] \cong I_n$. If $n = 8k$ with $k \geq 2$, then*

$$[q^{M_n - j}]P_{u, v} = 0 \quad (0 \leq j \leq k - 2).$$

If $n = 8k + 6$ with $k \geq 1$, then

$$[q^{M_n - j}]P_{u, v} = 0 \quad (0 \leq j \leq k - 1).$$

These assertions concern the top $k - 1$ and k allowed coefficients, respectively, without requiring any target cell or target a -function calculation.

Proof. For $n = 8k$ one obtains

$$M_n = 12k^2 - k - 1, \quad A_n = 12k^2 - 2k, \quad M_n - A_n = k - 1.$$

For $n = 8k + 6$ one obtains

$$M_n = 12k^2 + 17k + 5, \quad A_n = 12k^2 + 16k + 5, \quad M_n - A_n = k.$$

Every exponent listed in the corollary is strictly greater than A_n ; Theorem 2.1 therefore forces its coefficient to vanish. $\square$

n	L_n	$\ell(x_n)$	d_n	degree bound	μ
4	7	3	4	1	0
6	15	4	11	5	1
8	26	5	21	9	0
10	40	6	34	16	0
12	57	7	50	24	0
14	77	8	69	33	0
16	100	9	91	44	0
22	187	12	175	85	0
24	222	13	209	102	0

This table is arithmetic derived from the thesis formulas; it does not claim to recompute the high-rank polynomials. The special row $n = 8$ includes the improvement from Lemma 4.5.5.

Proposition 2.3 (Exact rank-eleven polynomial; computer assisted). *The project's exact recurrence computation gives*

$$F(q) := P_{x_6, w_6}^{D_6}(q) = 1 + 6q + 11q^2 + 6q^3 + q^4 + q^5.$$

Assuming the invariance input, every realization $[u, v] \cong I_6$ has $P_{u,v} = F$ and $P_{u,v}(1) = 26$.

Computational proof and transfer. The endpoints, with lengths four and fifteen, are

$$x_6 = (-1, -2, 4, 3, 6, 5), \quad w_6 = (-1, -6, 3, -4, 5, -2).$$

The saved Python implementation uses signed-inversion lengths, the Bruhat lifting property and the exact integer KL recurrence. Its separate R -polynomial recurrence checks

$$q^{11} P_{x_6, w_6}(q^{-1}) = \sum_{x_6 \leq z \leq w_6} R_{x_6, z}(q) P_{z, w_6}(q).$$

The closed interval has 1676 vertices and rank vector

$$(1, 12, 55, 140, 248, 339, 360, 287, 162, 59, 12, 1).$$

The lower ideal $[e, w_6]$ has 3184 vertices. The computation also checks each right-descent choice at w_6 ; its detailed recurrence certificate and verifier are saved with the project. The verifier never calls the KL evaluator: it checks 24,245 stored dependency records, including 17,148 strict recurrence identities and 6978 incomparable zero cases. All recurrence dependencies have shorter upper length, so these identities with the diagonal and zero base cases certify the answer by induction. This is a computer-assisted finite calculation, separate from the thesis's assertion of $\mu = 1$. The final transfer follows from the external invariance input. $\square$

Theorem 2.4 (Coefficients in abstract products). *Assume the invariance input. Suppose $[u, v]$ is isomorphic to $\prod_{i=1}^k [x_i, w_i]$, where each factor is a nontrivial type D interval with fully commutative x_i . Write $d_i = \ell(w_i) - \ell(x_i)$ and $d = \sum_i d_i$. Then*

$$P_{u,v}(q) = \prod_i P_{x_i, w_i}(q), \quad \deg P_{u,v} \leq \sum_i \left\lfloor \frac{d_i - 1}{2} \right\rfloor.$$

For $k \geq 2$ one has $\mu(u, v) = 0$. If every d_i is odd, the coefficient of $q^{(d-k)/2}$ is in $\{0, 1\}$. In particular, if $[u, v] \cong I_6^k$, then $P_{u,v} = F^k$, its degree is $5k$, and its actual leading coefficient is one.

Proof. The product poset is a Bruhat interval in the product Coxeter group. The Hecke algebra is the tensor product of the factor Hecke algebras; the tensor products of the canonical basis elements are bar invariant with the required triangular expansion, hence form the canonical basis. Their expansion gives the stated factorization of P . Invariance transfers it to $[u, v]$. The degree bound follows by adding factor bounds. Its right side is at most $(d - k)/2 < (d - 1)/2$ for $k \geq 2$, which implies the μ assertion. If all d_i are odd, reaching exponent $(d - k)/2$ requires the largest allowed exponent in every factor. That coefficient is $\prod_i \mu(x_i, w_i) \in \{0, 1\}$ by Gern's theorem. Finally F has degree five and leading coefficient one. $\square$

Tensor-product factorization is classical. The unrestricted transfer from an abstract product decomposition is the use of full invariance; no independent novelty claim is made for the product statement.

3 Certified small examples in types B and A

Use generators b_0, b_1, b_2 of B_3 , with b_0 negating the first signed entry, b_1 swapping the first two, and b_2 swapping the last two. The type D_4 interval

$$[(1, 3, 2, 4), (-1, 2, -3, 4)] = [s_3, s_3 s_1 s_2 s_3]$$

is isomorphic to the type B_3 interval

$$[(3, 2, -1), (2, -1, -3)].$$

Their endpoint lengths are $(1, 4)$ and $(4, 7)$, respectively; both have rank vector $(1, 4, 4, 1)$ and polynomial $1 + q$. The target bottom has reduced word $b_0 b_1 b_2 b_1$ and is not fully commutative, because $b_1 b_2 b_1$ is a noncommuting braid.

Here is the complete vertex bijection found by the computation:

relative rank	D_4 vertex	B_3 vertex
0	$(1, 3, 2, 4)$	$(3, 2, -1)$
1	$(-2, 3, -1, 4)$	$(3, -2, -1)$
1	$(2, 3, 1, 4)$	$(3, 1, -2)$
1	$(-3, -1, 2, 4)$	$(-3, 2, -1)$
1	$(3, 1, 2, 4)$	$(1, 2, -3)$
2	$(-1, 3, -2, 4)$	$(3, -1, -2)$
2	$(-3, 2, -1, 4)$	$(2, -3, -1)$
2	$(3, 2, 1, 4)$	$(2, 1, -3)$
2	$(-1, -3, 2, 4)$	$(-1, 2, -3)$
3	$(-1, 2, -3, 4)$	$(2, -1, -3)$

The verifier checks all pairwise order relations, not just rank numbers. It also checks every subinterval's KL polynomial, R -reciprocity, and every available right descent in the KL recurrence.

The B_3 interval is not contained in a left coset of a type A parabolic, even a conjugate one: two of its right cover reflections are $(-3, 2, -1)$ and $(-1, 2, 3)$, whose product has order four. Two distinct reflections in a type A Coxeter group have product order two or three; the cover labels of a coset of such a subgroup would lie in that subgroup. A corresponding pair of left cover labels of order four excludes right type A parabolic cosets as well. Thus the target example actually encounters the double-bond structure, although its KL equality is already covered by short-interval results.

Four additional saved certificates include B_4 intervals with non-fully-commutative bottom and ranks three and five, as well as type A_4 examples of those ranks. The rank-five examples among these four have $P = 1 + q$ and $\mu = 0$, illustrating the distinction between actual leading coefficient and μ . These small examples are verification demonstrations, not new results in those types.

A further rank-five certificate uses all four D_4 simple generators:

$$[(-2, -1, 3, 4), (-4, -3, -2, -1)]_{D_4} \cong [(2, 4, 3, -1), (2, 3, -1, -4)]_{B_4}.$$

The endpoint lengths are $(1, 6)$ and $(5, 10)$. Each interval has 34 vertices. Its rank vector and polynomial are

$$(1, 4, 10, 12, 6, 1), \quad P = 1 + 2q + q^2, \quad \mu = 1.$$

The D_4 bottom is s_1 , and a reduced word of the top is $s_1 s_3 s_2 s_4 s_3 s_1$. The B_4 bottom has reduced word $b_0 b_1 b_2 b_3 b_2$, containing the braid $b_2 b_3 b_2$. The source's top has full D_4 support, and the target also has saved order-four cover-reflection obstructions to type A cosets. This is a useful example with a fork in the source and double bond in the target; its rank remains within previously known invariance.

4 A structural obstruction and the shape of I_6

Proposition 4.1 (Eight-element dihedral obstruction). *Let $K = [e, w_0]$ be the principal interval in $I_2(4) = B_2$, with level sizes $(1, 2, 2, 2, 1)$. No interval in a finite simply laced Weyl group is isomorphic to K . Consequently, any interval containing a closed subinterval isomorphic to K has no finite type D realization, regardless of its bottom's FC status.*

Proof. We give the classical reflection-plane argument; see Dyer [14, Lemma 3.1 and Section 3.4] for its antecedents. Label a cover $z < z'$ by the reflection $z'z^{-1}$ and its positive root. In a diamond with bottom z , middle vertices $r_\alpha z, r_\beta z$, and top $r_\rho r_\alpha z = r_\sigma r_\beta z$, one has

$$r_\rho r_\alpha = r_\sigma r_\beta.$$

In a finite real reflection representation, a nonidentity product of two distinct reflections has $\text{im}(r_\rho r_\alpha - 1) = \text{span}(\rho, \alpha)$. The same image is $\text{span}(\sigma, \beta)$. The first two roots α, β are distinct and independent, so all four diamond labels lie in their common plane.

In K , every vertex at one of the three middle levels is covered by both vertices at the next middle level, when that level exists. Apply the diamond argument first from the bottom to each rank-two vertex. All labels at ranks zero through two lie in one plane. Apply it next to diamonds from rank one to rank three, and then from rank two to the top. At each step the two already known outgoing reflections are distinct, so their roots span that same plane. All cover labels consequently lie in that plane.

The eight vertices must then lie in one coset of the reflection subgroup generated by roots in this plane: reach each vertex by a cover path from the bottom. In a finite simply laced root system, such a rank-two subsystem is $A_1 \times A_1$ or A_2 , whose reflection groups have four or six elements. A coset cannot contain eight distinct vertices, a contradiction. The last assertion follows by restricting any proposed interval isomorphism to the specified closed subinterval. $\square$

Since e is FC in B_2 and $P_{e,w_0} = 1$, this obstruction is stronger than a test using $\mu \geq 2$. In particular, it disproves the proposed shortcut that every FC-bottom interval in every finite type can be modeled by a finite- D FC-bottom interval. It is a classical structural obstruction, not a new result of this project.

Proposition 4.2 (Shape certificates for I_6 ; computer assisted). *The interval I_6 is not isomorphic to a principal lower interval in any Coxeter system, and it is not a lattice.*

Proof. The saved Bruhat-order computation gives rank eleven and twelve atoms. A principal interval $[e, z]$ of rank eleven has at most eleven atoms: they are the distinct simple generators occurring in a reduced expression of z , which has eleven letters. This proves the first assertion.

For the second, take the following two atoms of I_6 :

$$a = (-4, -2, 1, 3, 6, 5), \quad b = (-1, -3, 4, 2, 6, 5).$$

Their two distinct minimal common upper bounds are

$$c = (-4, -3, 1, 2, 6, 5), \quad d = (-2, -4, 3, 1, 6, 5).$$

They have relative ranks two and three, respectively, and are incomparable. The saved script enumerates all 720 common upper bounds, verifies c, d are exactly the minimal ones, and checks the comparisons using the Bruhat lifting property. Thus a, b have no join, so I_6 is not a lattice. This order-only certificate does not use a KL-polynomial calculation. $\square$

5 Concrete other-type families and why they are classical

The transfer hypotheses can be met in other types without any new invariance theorem. The following explains both a useful application and its novelty limitation.

Lemma 5.1 (Convex parabolic cosets). *Let $J \subseteq S$, let W_J be its standard parabolic subgroup, and let $a \in W^J$ be the minimal element of aW_J . For $x \leq w$ in W_J , multiplication by a identifies $[x, w]_{W_J}$ with the entire interval $[ax, aw]_W$.*

Proof. Parabolic decomposition $z = z^J z_J$ is length-additive, and the projection $z \mapsto z^J$ is order preserving. If $ax \leq z \leq aw$, then $a \leq z^J \leq a$, hence $z \in aW_J$. On this coset, $az_1 \leq az_2$ if and only if $z_1 \leq z_2$ in W_J . These are the standard parabolic Bruhat-order facts; see [7, Section 2.5]. They identify precisely the vertices and order of the two intervals. $\square$

Proposition 5.2 (Classical coset polynomial equality). *Under the hypotheses of Lemma 5.1, the equality*

$$P_{ax,aw}^W = P_{x,w}^{W_J}$$

holds independently of the invariance input.

Proof. Induct on $\ell(w)$. The case $x = w$ is immediate. Choose a right descent $s \in J$ of w . Length additivity shows that the right descents in J of ax and aw are the corresponding descents of x and w . Use the standard right-handed KL recurrence with s . Its first two terms have upper element $aws = a(ws)$, so the induction hypothesis applies (and incomparable terms are zero). Every nonzero summand in its correction sum has $ax \leq z \leq a(ws)$ and $zs < z$. Lemma 5.1 puts $z = ay$ with $x \leq y \leq ws$ in W_J . The descent condition and all length differences match. The polynomial and μ factors in that sum have smaller upper parabolic length and match by induction. Thus this recurrence is exactly the recurrence in W_J . The recurrence is Gern [1, Proposition 1.2.6], with left and right exchanged, or [7, Section 6.1]. $\square$

For example, take an irreducible affine Weyl group $\tilde{D}_n$ and its finite standard D_n parabolic J . With $\rho^\vee$ half the sum of the positive coroots, let

$$a_m = t_{-2m\rho^\vee}, \quad m \geq 1.$$

Since $2\rho^\vee$ lies in the coroot lattice, these are affine Weyl group elements. Use the affine-root convention $t_\lambda(\alpha + r\delta) = \alpha + (r - \langle \lambda, \alpha \rangle)\delta$. For finite simple α_i , $a_m(\alpha_i) = \alpha_i + 2m\delta$ is positive; thus $a_m \in W^J$. Conversely, $a_m^{-1}(\alpha_i) = \alpha_i - 2m\delta$ is negative, so every finite simple generator lies in the left descent set of a_m . An adjacent pair of finite simple generators consequently forces a reduced prefix sts , with $m(s, t) = 3$, in a_m by parabolic decomposition. Therefore a_m and its length-additive extensions a_mx_n are non-fully-commutative. This descent argument is also Gern [1, Lemma 1.1.21 and Corollary 1.1.22]. Affine translations and the root-sign length criterion are standard; see [8, Chapters 4 and 5].

Lemma 5.1 and Proposition 5.2 now give

$$[a_mx_n, a_mw_n] \cong I_n, \quad P_{a_mx_n, a_mw_n}^{\tilde{D}_n} = P_{x_n, w_n}^{D_n}.$$

In particular, $\mu = 1$ for $n = 6$ and $\mu = 0$ for the other even values, with all the degree bounds in Theorem 2.1. The endpoint lengths tend to infinity as m tends to infinity. This gives explicit affine examples with non-fully-commutative bottom, but Proposition 5.2 makes their equality classical. They cannot be presented as evidence that full invariance was needed.

Finite E_7 and E_8 contain standard D_6 parabolics, and E_6 contains a D_4 parabolic. Their standard parabolic intervals, and coset copies as above, provide analogous exceptional-type examples. Again these are already covered by parabolic theory, and do not settle the fully commutative-bottom bound for all of an exceptional type.

For fully explicit rank-eleven exceptional examples, use the numbering 1–3–4–5–6–7 (and 7–8 in E_8), with 2 attached to 4. Set $J = \{2, 3, 4, 5, 6, 7\} \cong D_6$ and relabel Gern’s generators $(1, 2, 3, 4, 5, 6)$ as $(2, 3, 4, 5, 6, 7)$. Let x, w be the resulting x_6, w_6 , and put $a = w_0(E_r)w_0(J)$ for $r = 7, 8$. This is the minimal element of aW_J , of length 33 in E_7 and 90 in E_8 . Consequently (ax, aw) has lengths $(37, 48)$ and $(94, 105)$, respectively, and polynomial F from Proposition 2.3. The saved exact integer-root script checks that both lower endpoints are non-fully-commutative: stripping a displayed sequence of right descents from a leaves two adjacent descents, hence a reduced noncommuting braid. Concatenation with a reduced word for x preserves that braid. The script and application notes contain the full sequences and root images. These particular pairs are classical coset applications.

6 Literature and novelty audit

The audit is a documented search through October 6, 2026, not a proof of the absence of every unpublished or inaccessible result.

Candidate statement	Assessment and primary reference
Finite type A FC-bottom 0–1 bound	Known: Green [3, Theorem 4.1; arXiv Theorem 3.1]. His result also covers affine type A .
Finite type D FC-bottom bound	The supplied thesis: Gern [1, Theorem 4.5.11].
Finite type B/C FC-bottom bound	Known: Chmutov’s 2014 thesis [4, Theorem 2.3.8]. Types B and C have the same ordinary equal-parameter Coxeter system.
Both endpoints FC in B and H	Known: Green [15, Remark 7.13]. This does not alone prove the FC-bottom/arbitrary-upper statement.
Upper endpoint Deodhar	Known: Jones [16, Theorem 1.2], including exceptional finite Weyl types. This uses a different endpoint hypothesis; the bad elements w_n are not FC.
Subsystem interval pattern transfer	Known: Woo [6, Corollary 3.3]; the arXiv numbering is Corollary 2. Those embeddings already preserve KL polynomials.
Principal-lower-interval transfers	Known: Brenti–Caselli–Marietti [10, Theorem 7.8 and Corollary 8.4] and Delanoy [11].
Small arbitrary intervals	Known through rank six for arbitrary Coxeter systems: Barkley–Gaetz–Lam [12, Corollary 1.4].
Small finite Weyl/type A intervals	Known through rank ten for Weyl groups and rank twelve when both realizations have type A : Esposito–Marietti–Stella [13, Corollary 1.3(ii)].
Arbitrary realizations of I_n	Theorem 2.1 is a formal consequence of full invariance and Gern’s calculations. Its unrestricted scope exceeds these short-interval results, but an independent novelty claim is not established.

The rank-eleven I_6 test and rank-twenty-one I_8 test should not be described as outside *every* prior result merely because their ranks exceed ten. A particular realization may be handled by special matchings, geometry, parabolic cancellation, or pattern embedding. Nor does the rank-twelve type A statement automatically compare a type A realization with a type D realization; its precise ambient hypotheses matter. The defensible claim here is the new unrestricted *deduction*, conditional on the supplied full theorem. For I_6 , the shape certificate above specifically rules out principal-lower and lattice realizations. It does not rule out every other older transport mechanism for a particular realization.

7 Verification, computations, and remaining research

The project saves the source papers and extracted text, a literature audit, and executable Python code under `computations/`. The finite verification exhaustively compares two implementations on 54,907 comparable pairs (including equal endpoints) and 201,600 ordered Bruhat comparisons in A_3, A_4, B_3, D_4, B_4 . They use BFS versus inversion-formula lengths and subword versus lifting-property Bruhat comparison; both implement the same standard KL recurrence. The independent R recurrence adds a different polynomial identity check for the certified examples. The saved JSON report includes six complete order-isomorphism certificates and the targeted D_6 result. The D_8 base computation is inherited from the thesis and was not independently reproduced. The executed dependency-free command is

```
python3 computations/run_search.py
```

It can also be run through `uv run -offline` using an installed Python; project instructions give the exact command and certificate verification procedure. The exceptional root checks run with

`python3 research/exceptional_cosets.py`. Arithmetic degree tables, small-group KL calculations, and theorem proofs have different evidentiary roles: the table checks formulas; small-group calculations check concrete examples; neither proves unrestricted combinatorial invariance or whole-type coverage.

The logical audit of Theorem 2.1 is short and exact:

1. The interval certificate supplies the whole closed poset, hence its rank.
2. The external invariance theorem identifies the two ordinary polynomials.
3. Gern’s Lemma 2.3.9 permits the e -bottom a -function degree bound.
4. Proposition 4.5.1 evaluates that source a -function.
5. Taking integer floors and the ordinary degree bound yields B_n .
6. Lemma 4.5.5 supplies the original computational base values; our certificate independently reproduces and expands the $n = 6$ calculation. The $n = 8$ value remains inherited from the thesis and removes the allowed q^{10} term.

No target a -function, target fully commutative status, or target root-label compatibility is assumed. No assertion is made for unequal-parameter polynomials, parabolic variants, or p -Kazhdan–Lusztig polynomials.

Update from the expanded search. The companion paper *Leading Kazhdan–Lusztig coefficients with fully commutative lower endpoints in type E* gives a computer-assisted extension to entire finite types E_6, E_7, E_8 . Independent parabolic searches certify E_8 without enumerating its full group. The note also proves explicit unbounded full-support upper families in affine E_8 and indefinite E_{10}, E_{13} . The affine family has zero μ for every FC lower endpoint; the indefinite families have the FC-bottom 0/1 bound. Classical cell theory and parity supply the structural proof. These direct results enlarge the conditional abstract model class; they do not provide new interval isomorphisms. Their historical priority remains unconfirmed. The expanded search also independently verifies a fully commutative-bottom affine- D_4 pair with $\mu = 2$: with central node 2 and leaves 0, 1, 3, 4, put $x = s_0 s_1 s_3 s_4$ and $w = x s_2 x$. Then $P_{x,w} = 1 + 3q + 2q^2$. The cube-interval interpretation is classical, and Mongelli’s 2011 preprint *Kazhdan–Lusztig polynomials of boolean elements*, Corollary 2, explicitly covers the star pattern and its Catalan leading coefficient. This is a rediscovery, not a novelty claim.

Large μ -values in affine types B, C, D are known [17]; they are not counterexamples to an FC-bottom assertion without checking their lower endpoints. Likewise the initial literature search’s failure to find a complete exceptional-type or affine- D FC-bottom theorem is not a certification that such a theorem remains open. The separate literature file records queries, exact locators, version differences, and access limitations, including the university repository’s blocked Chmutov PDF download.

For a substantive next theorem, the missing step is a structural recognition result supplying FC-bottom D witnesses for a naturally defined family in exceptional or affine type *beyond* classical parabolic or subsystem embeddings. Another promising target is a certified non-parabolic realization of I_6 or I_8 . Such a high-rank nonclassical realization has not been proved. The exceptional companion supplies separate direct finite-type and explicit infinite-family extensions. Its rank-uniform construction also gives full-support upper families in every E_{4r+1} , $r \geq 3$, with a second unbounded parameter. Their FC-bottom leading coefficients are 0 or 1, and all non-cover coefficients vanish. A

symbolic D_{4r} sign-change proof establishes real-root membership; no KL polynomial computation is needed. Historical priority remains unconfirmed, and full polynomials are not determined. This note records completed conditional deductions and concrete classical applications without replacing those gaps by an unproved novelty claim.

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